

Experiment 4: Deflections of beams

Objectives

Understand the behavior of beams under bending loads, and testing the predictions of Euler-Bernoulli beam theory.

Background

Beams are slender bodies that resist external loads primarily by way of bending. They can commonly be found as structural members supporting slabs in buildings. Horizontal members supporting the top surface of your table, or chair, also qualify as beams.

The most common theory in practice to predict the deflections, and consequently stresses, in beams is the Euler-Bernoulli beam theory. In this experiment, we will put this theory to test by comparing it to the Euler-Bernoulli beam theory.

Before conducting the experiment, we must first answer the following questions:

- 1.) What are the assumptions on the deformation of a beam that the Euler-Bernoulli theory makes?
- 2.) For a beam with Young's modulus E and second moment of area I , loaded with a uniformly distributed load q along its length, the deflection y from a straight unloaded configuration is governed by the equation:

$$\frac{d^4 y}{dx^4} = \frac{q}{EI}$$

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 - 3.) Use the above equation to obtain a formula for the deflected shape, i.e. $y(x)$, of a cantilever beam, with a point load P applied at the free end.
- Use the deflected shape to compute expressions for bending moment and shear force in the cantilever beam.

Some online resources

You may watch the following resources to refresh your memory on bending of beams

[Beam bending \(kinematics\)](#) (1:03 minutes)

[Deflections of beams](#) (22:14 minutes)

The experiment

Task 1:

The apparatus to load beams and measure their deflections is available in the ME lab. Your job is to setup a cantilever beam with 5 different values of point load applied at its free end. The value of the loads must be chosen such that the **deflection of the free end remains small (say less than 3% of the total length)**. For each value of the applied load, measure the deflection of the beam at distances $0.25L$, $0.5L$, $0.75L$, and L from the fixed end. Take help from the lab staff and TAs if you need to. Compare these deflections with the theoretical formula of the deflected shape obtained using the Euler-Bernoulli beam theory in the “Background” section above.

Task 2:

Write a report on the entire exercise. The report must not exceed 3 pages. You can tailor your report as per the following proposed structure, but you are free to deviate from it if you wish so.

- 1.) **INTRODUCTION:** State your objectives, and the answers to questions 1-3 stated in the “Background” section above.
- 2.) **PROCEDURE:** Describe the details of the procedure used to conduct the experiment.
- 3.) **OBSERVATIONS:** Report all the experimental data recorded. Describe your observations, insights, any calculations you perform. Use plots and tables to compare experiments with theory. Does your experimental data agree with the theoretical predictions? If not, explain the possible reasons.
- 4.) **CONCLUSIONS:** Describe what you understood from this entire exercise.